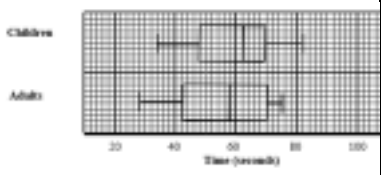
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	86	**2**	**M1** for correct method to find the perimeter e.g. $(8 + 3) \times 2 \times 5 - 3 \times 8$ If 0 scored, **SC1** for answer 98
2	15	**1**	
3(a)	66	**1**	
3(b)	Positive	**1**	
3(c)	Ruled line of best fit	**1**	
3(d)	46 to 50	**1**	**FT** *their* line of best fit if a positive gradient
4(a)	0.22 oe	**2**	**M1** for $0.15 + 0.2 + ? + 0.43 = 1$ or better
4(b)	40	**1**	
5(a)	52	**1**	
5(b)	$7n + 5$ oe final answer	**2**	**B1** for $7n + a$ or $bn + 5$ $b \neq 0$
6	7	**3**	**M2** for $166 + 2x = 180$ or better or **M1** for $97 - 3x + 69 + 5x = 180$ oe
7	$2^5 \times 3^4 \times 13^2$	**1**	
8	$\frac{56}{24} - \frac{21}{24}$	**M2**	**M2** for correct method for common denominator or **B1** for $\frac{7}{3}$
	their $\frac{35}{24} \times \frac{6}{25}$	**M1**	
	$\frac{7}{20}$	**A1**	
9(a)	$7a(3a + 4b)$ final answer	**2**	**B1** for partial factorisation $7(3a^2 + 4ab)$ or $a(21a + 28b)$

Question	Answer	Marks	Partial Marks
9(b)	$5(2x + 3y)(2x – 3y)$ final answer	**3**	**B2** for $(2x + 3y)(2x – 3y)$ or $(10x + 15y)(2x – 3y)$ or $(2x + 3y)(10x – 15y)$ or **B1** for $5(4x^2 – 9y^2)$
10	[x =] 55 [y =] 24	**2**	**B1** for each
11	990	**3**	**M2** for correct complete area statement e.g. $\frac{1}{2} \times 30 \times (6 + 12) + 60 \times 12$ oe or **M1** for one area calculation
12(a)	22	**2**	**B1** for 48 and 70
12(b)		**2**	**M1** for a box with two whiskers and at least two correct from Min 28, LQ 42, Med 58, UQ 70, Max 75
13	16.6 or 16.64…	**5**	**M2** for $21 \times \frac{18}{13.5} = [AC]$ oe or **M1** for scale factor $\frac{13.5}{18}$ or $\frac{18}{13.5}$ oe soi Then Pythagoras method: and **M2** for $\sqrt{28^2 + 18^2}$ [÷ 2] or $\sqrt{(theirAC)^2 + 18^2}$ [÷ 2] or **M1** for $AD^2 = 28^2 + 18^2$ or $AD^2 = (theirAC)^2 + 18^2$ OR alternative trigonometry method e.g. **M1** for $\tan E = \frac{21}{13.5}$ and **M1** for $AD = \frac{18}{\cos their\, 57.3}$
14(a)	[p =] –13	**2**	**M1** for $4(5x – 4) + 3$ or better
14(b)	$\frac{3x+1}{5}$	**3**	**M2** for $x = \frac{3y+1}{5}$, $5y = 3x + 1$ or $y - \frac{1}{5} = \frac{3x}{5}$ **M1** for $x = \frac{5y-1}{3}$, $3y = 5x - 1$ or $y + \frac{1}{3} = \frac{5x}{3}$